UNIVERSITÉ DE GENÈVE
FACULTY OF SCIENCE
Department of Informatics

stochastic information processing

Quantum computing

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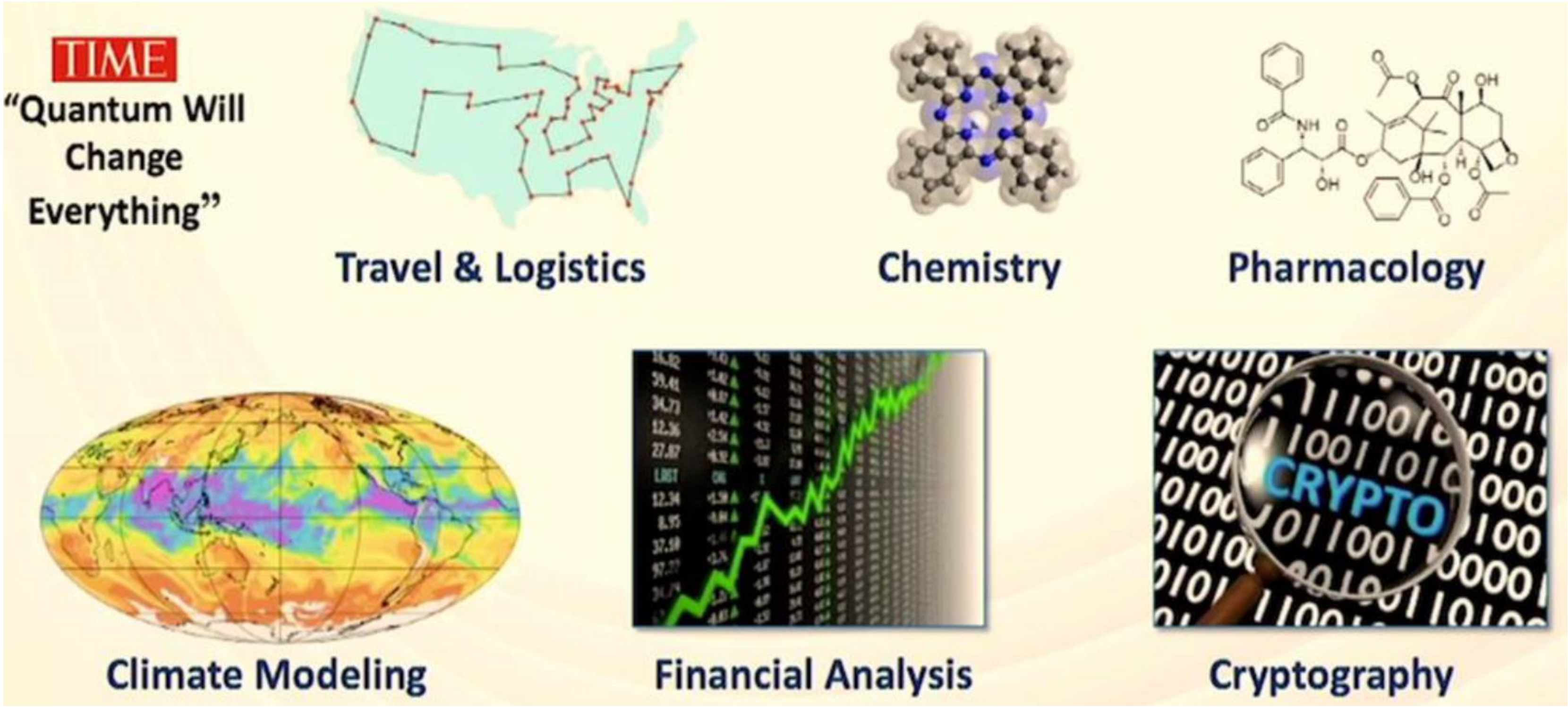
Content of course

- Theoretical basis for quantum computing.
- Practical basis for quantum computing.
- Near-term potential of quantum computing.
- Other applications of quantum information.

Applications of quantum computers



Applications of quantum computers

Application 1. Quantum Factoring

P. Shor (1994)

A quantum computer can factor numbers **exponentially faster** than classical computers

15 = 3 × 5 (...easy)
38647884621009387621432325631 = ? × ?

Importance: cryptanalysis

public key cryptography relies on inability to factor large numbers

Best classical algorithm: 10^{24} steps	Shor's quantum algorithm: 10^{10} steps
On classical THz computer: 150,000 years	On quantum THz computer: <1 second

Applications of quantum computers

Application 2: Quantum Search

L. Grover (1997)

A quantum computer can find a marked entry in an unsorted database **quadratically faster** than classical computers

(e.g., given a phone number, finding the owner's name in a phonebook)

Importance: "satisfiability" problems

- fast searching of big data
- inverting complex functions
- determining the median or other global properties of data
- pattern recognition; machine vision

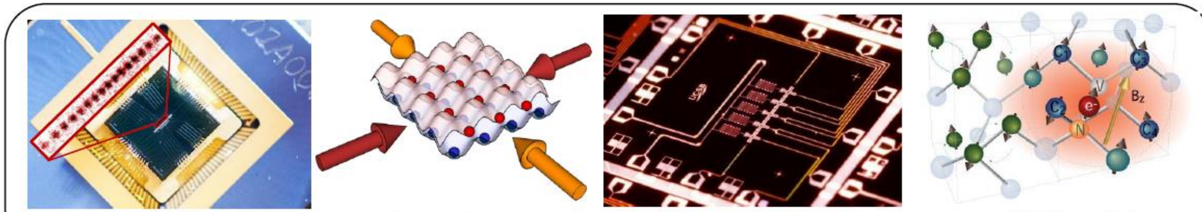
Applications of quantum computers

Application 3: Quantum Simulation

Quantum modelling is hard: *N* quantum systems require solution to 2^N coupled eqns

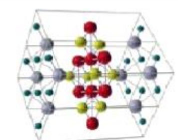
$$i\hbar \frac{\partial \Psi}{\partial t} = H\Psi$$

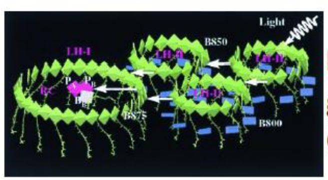
Alternative approach: Implement model of interacting system on a ***quantum simulator***, or "standard" set of qubits with programmable interactions



Atomic ions | Trapped atoms | Superconductors | NV-diamond

Quantum Material Design Understand exotic material properties or design new quantum materials from the bottom up





Energy and Light Harvesting Use quantum simulator to program QCD lattice gauge theories, test ideas connecting cosmology to information theory (AdS-CFT etc..)

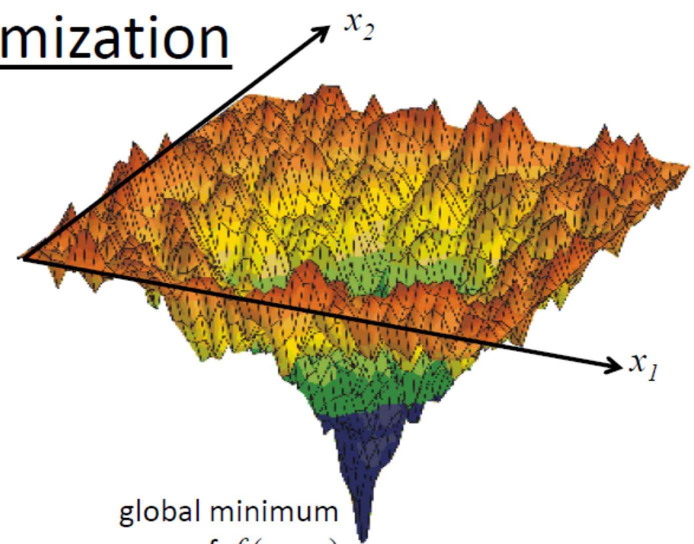
Quantum Field Theories Program QCD lattice gauge theories, test ideas connecting cosmology to information theory (AdS-CFT etc..)

Applications of quantum computers

Application 4: Quantum Optimization

Minimizing complex (nonlinear) functions by "simultaneously sampling" entire space through quantum superposition

Relevant to
- Logistics
- Operations Research
- VLSI design
- Finance



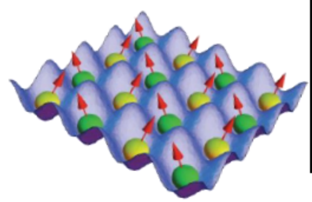
global minimum of $f(x_1,x_2)$

Example: quadratic optimization

Minimize

$$f(x_1, x_2, \dots) = \sum_{i<j} q_{ij} x_i x_j + \sum_i c_i x_i$$

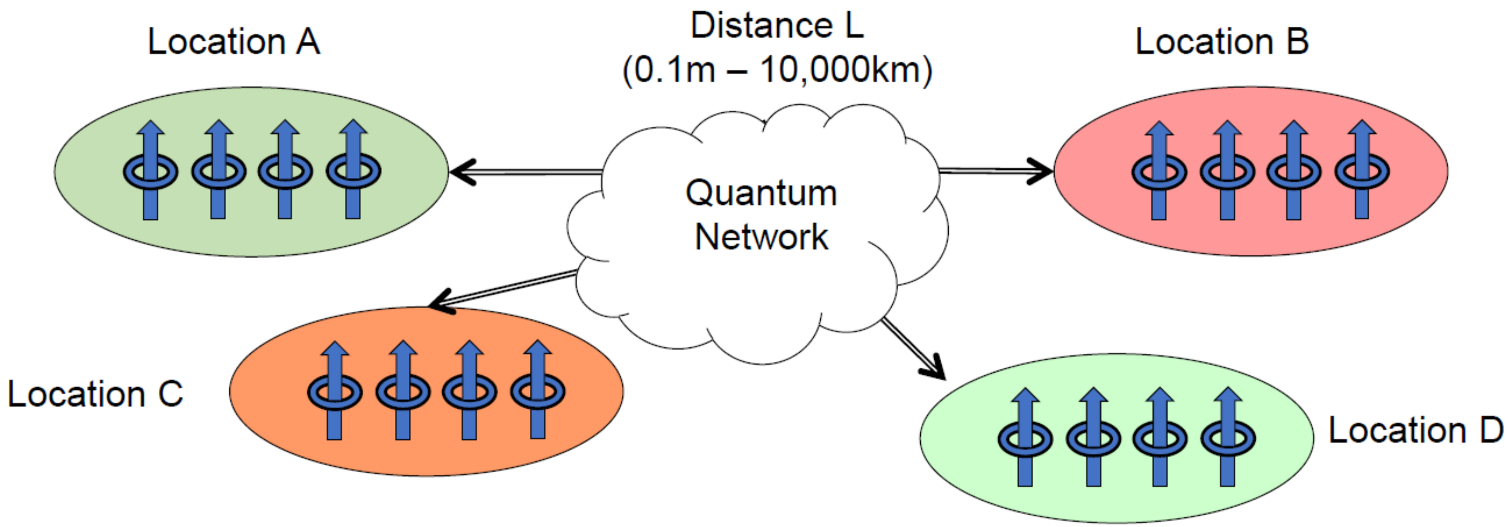
this function maps to energy of quantum magnetic network



Killer Application?
- could crack a large class of intractable problems: factoring, "traveling salesman" problem, etc..
- BUT not known if there is always a quantum speedup

Applications of quantum computers

Application 5: Quantum Networks



Uses of a quantum network

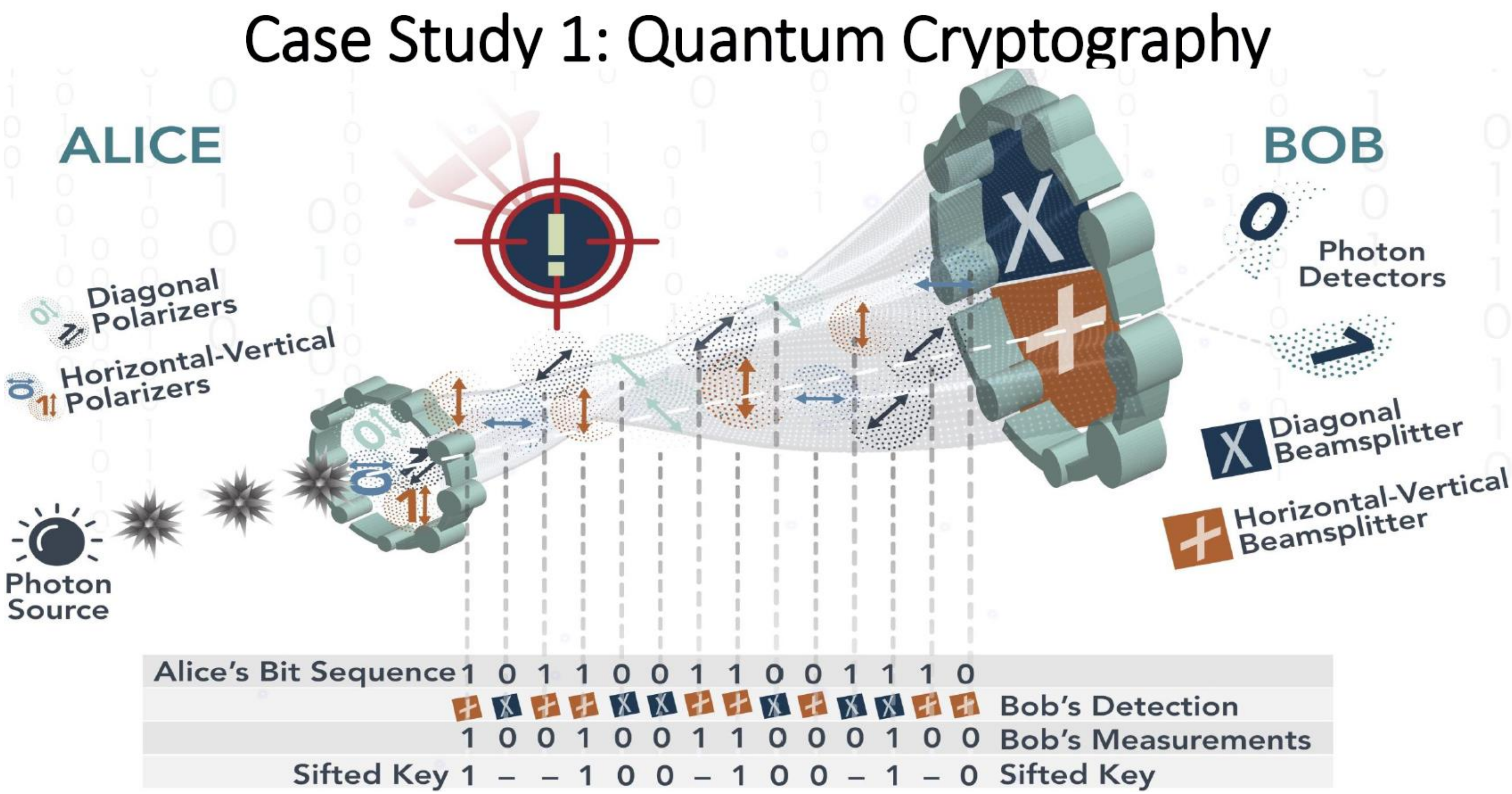
- Secret key generation: cryptography
- Certifiable random number generation
- Quantum repeaters ("amplifiers")
- Distributed quantum entanglement for optimal decision making
- *Large-scale quantum computing*

Applications of quantum computers

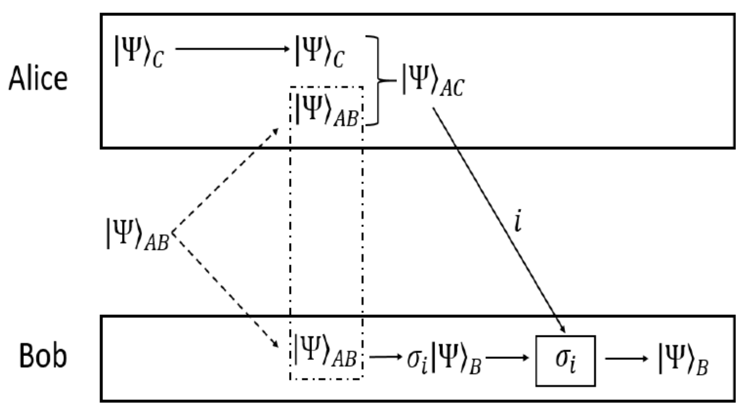
Case Study 1: Quantum Cryptography

- Alice and Bob wish to establish a secret key, but the communication lines between them may be compromised. Alice sends random q-bits to Bob. For each one, she picks either basis $\{|0\rangle, |1\rangle\}$ or $\{|0\rangle \pm |1\rangle\}$.
- Bob randomly chooses a basis to measure the bit in. If he guesses wrong, he gets a random bit.
- If an eavesdropper Eve tries to intercept the q-bits, she doesn't know (any more than Bob) the basis in which they were prepared. If she guesses wrong, the bit she passes on to Bob will have been disturbed.
- After N bits have been sent, Alice and Bob compare notes as to which bases they used. All bits where their bases didn't match are discarded. The remainder should be perfectly correlated.

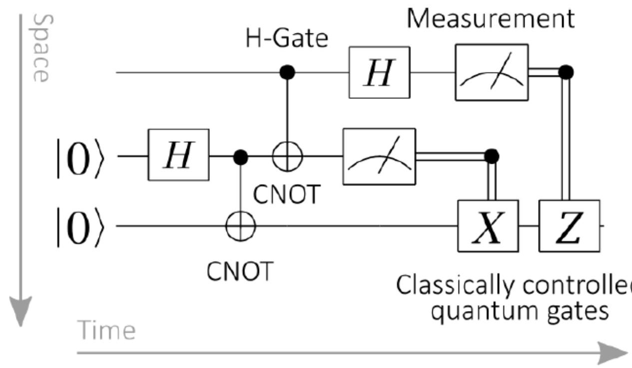
Applications of quantum computers



Applications of quantum computers



- Alice's measurement disentangles A and B and entangles A and C. Depending on what particular entangled state Alice sees, Bob will know exactly how B was disentangled, and can manipulate B to take the state that C had originally. Thus, the state C was "teleported" from Alice to Bob, who now has a state that looks identical to how C originally looked.
- It is important to note that state C is not preserved in the processes: the **no-cloning** and **no-deletion** theorems of quantum mechanics prevent quantum information from being perfectly replicated or destroyed.
- Bob receives a state that looks like C did originally, but Alice no longer has the original state C in the end, since it is now in an entangled state with A.



Quantum Teleportation Explained

1) Generate an entangled pair of electrons with spin states A and B, in a particular Bell state
2) Separate the entangled electrons, sending A to Alice and B to Bob.
3) Alice measures the "Bell state" (described below) of A and C, entangling A and C.
4) Alice sends the result of her measurement to Bob via some classical method of communication.
5) Bob measures the spin of state B along an axis determined by Alice's measurement

Applications of quantum computers

Case Study 2: Shor Algorithm of Factoring

- Theorem of Number Theory
  - If $a \neq \pm b$ (mod N) but $a^2 \equiv b^2$ (mod N), then gcd(a+b,N) is a factor of N.

- To Factor N on a quantum computer:
  - Select x coprime to N
  - Use QFT on quantum computer to find the period of
    $$f(s) = x^s \text{ mod } N$$
  - Use order of x to compute possible factors of N.
  - Check if they work; If not, rerun.
- Best classical algorithm takes time $O(\exp(n^{1/3}))$

  But Shor's quantum algorithm takes time $O(n^3 \log n)$

Peter Shor 1994

Applications of quantum computers

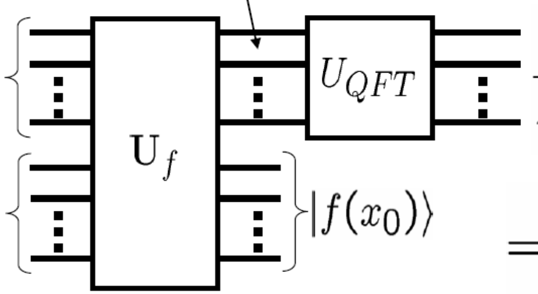
Quantum Fourier Transform (QFT)

$$U_{QFT} = \frac{1}{\sqrt{N}} \sum_{y=0}^{N-1} \sum_{x=0}^{N-1} \omega_N^{xy} |y\rangle\langle x| \qquad U_{QFT}^{\dagger} = \frac{1}{\sqrt{N}} \sum_{y=0}^{N-1} \sum_{x=0}^{N-1} \omega_N^{-xy} |x\rangle\langle y|$$

$$|\psi(x_0)\rangle = \frac{1}{\sqrt{2^n/l}} \sum_{j=0}^{2^n/l-1} |x_0 + jl\rangle$$

$$\frac{1}{\sqrt{2^n}} \sum_{x=0}^{2^n-1} |x\rangle$$

$$|0\rangle$$



$$|f(x_0)\rangle$$

$$\frac{1}{\sqrt{2^{2n}/l}} \sum_{p=0}^{2^n-1} \sum_{j=0}^{2^n/l-1} \exp\left(\frac{2\pi i p(x_0 + jl)}{2^n}\right) |p\rangle$$

$$= \frac{1}{\sqrt{l}} \sum_{p=0|pl=0 \text{ mod} 2^n}^{2^n-1} e^{\frac{2\pi i p x_0}{2^n}} |p\rangle$$

$$f(s) = x^s \text{ mod } N$$

Applications of quantum computers

Case Study 2: Shor Algorithm of Factoring

1881988129206079638386972394616504398071635633794173827007633564229888597152346654853190606065047430453173880113033967161996923212057340318795506569962213051687593076502570 59

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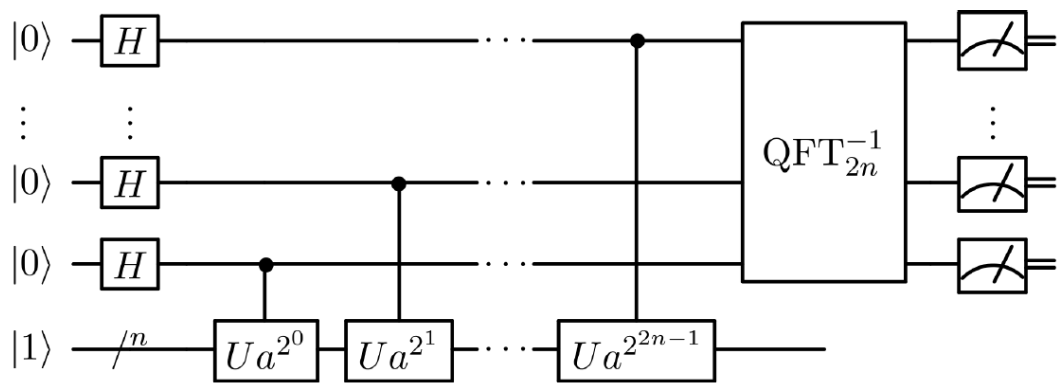


Applications of quantum computers

Case Study 3: Variational Quantum Eigensolver

- VQE is a quantum/classical hybrid algorithm that can be used to find eigenvalues of a matrix H.
- When VQE is used in quantum simulations, H is typical the Hamiltonian of an electronic system.
- It is run inside of a classical optimization loop.
- The quantum part has two fundamental steps:
  - Prepare the quantum states $|\psi\rangle$, often called the ansatz;
  - Measure the expectation value $\langle\psi|H|\psi\rangle$
- The classical optimization loop does two tasks:
  - Use a classical non-linear optimizer to minimize the expectation value by varying ansatz parameters
  - Iterate until convergence is reached.

Applications of quantum computers

Post-Quantum World

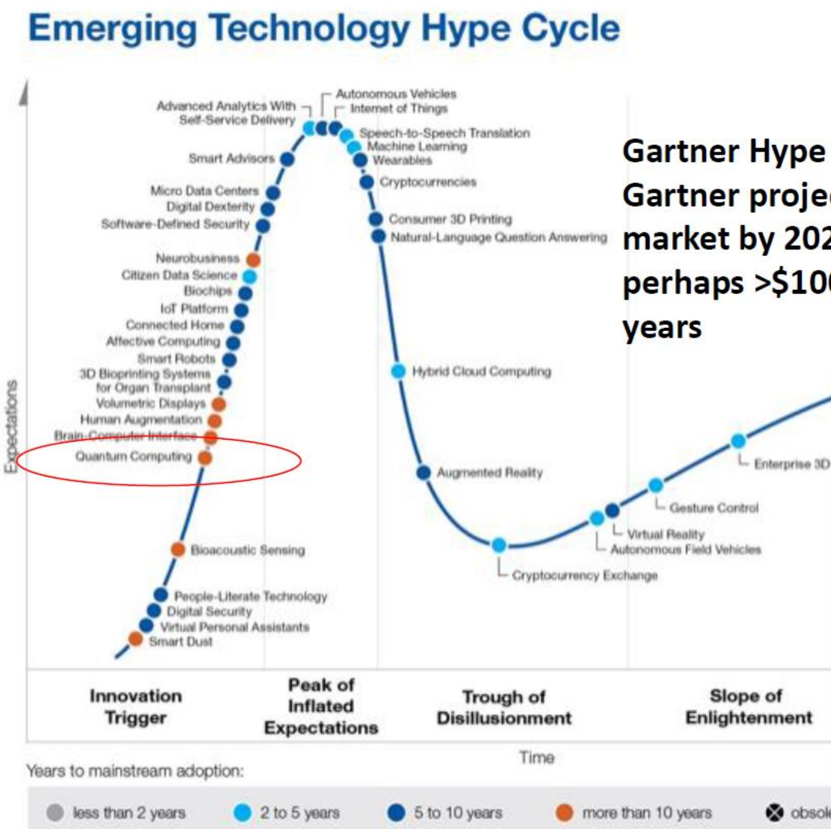
There is a good chance that quantum computers will be able to crack RSA-2048 within five-ten years (need about 8,000 qubits in a universal computer to do this).
Some encrypted data has a shelf-life of more than ten years. It may take ten years to cut over to a new encryption scheme, so companies and governments are scrambling to figure out what to do.

Most encryption used today is not safe in a quantum world.
If someone has been recording a https session, say, they may not be able to decrypt it now, but a few years from now, who knows.

There are encryption algorithms that are safe:
Secure communications will move to symmetric encryption or hash algorithms, or perhaps lattice algorithms.

QKD (Quantum Key Distribution) will likely be more popular for the most sensitive communications
Some governments and banks are already starting to use “Quantum Safe” algorithms, and this is expected to increase over the next decade.

Applications of quantum computers

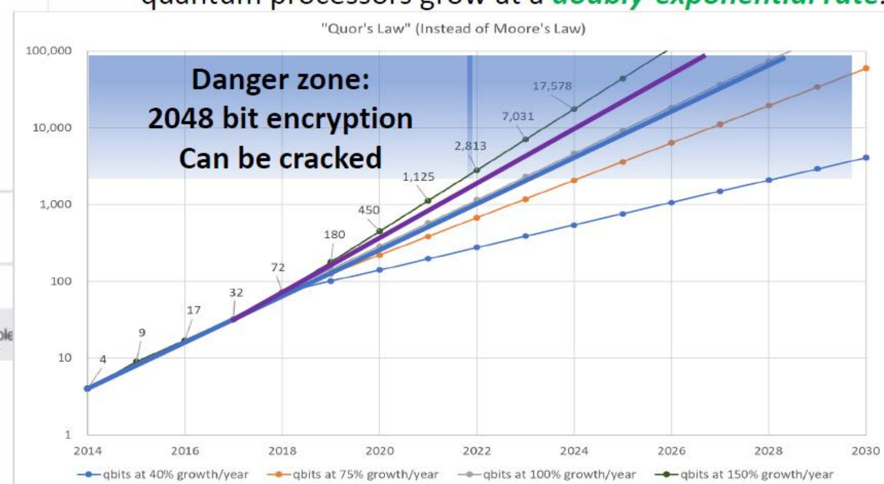


Gartner Hype Cycle. Gartner projects a $5B market by 2020 worth perhaps >$100B in ten years

Future of Quantum Computers

Qubit count of general-purpose quantum computers will double roughly every one to two years, or roughly every 18 months on average.

Neven's Law, named after Hartmut Neven, the director of Google's Quantum Artificial Intelligence Lab, stated that quantum processors grow at a ***doubly-exponential rate***.



Applications of quantum computers

Problems with Quantum Computers

- Decoherence
  - Quantum system is extremely sensitive to external environment, so it should be safely isolated
  - It is hard to achieve the decoherence time that is more than the algorithm running time
- Error correction (requires more qubits!)
- Physical implementation of computations
- New quantum algorithms to solve more problems
- Entangled states for data transfer
- Running time increases exponentially with matrix size